

# Pathogen exploitation of an abscisic acid- and jasmonate-inducible MAPK phosphatase and its interception by *Arabidopsis* immunity

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Phytopathogens promote virulence by, for example, exploiting signaling pathways mediated by phytohormones such as abscisic acid (ABA) and jasmonate (JA). Some plants can counteract pathogen virulence by invoking a potent form of immunity called effector-triggered immunity (ETI). Here, we report that ABA and JA mediate inactivation of the immune-associated MAP kinases (MAPKs), MPK3 and MPK6, in *Arabidopsis thaliana*. ABA induced expression of genes encoding the protein phosphatases 2C (PP2Cs), *HAI1*, *HAI2*, and *HAI3* through ABF/AREB transcription factors. These three HAI PP2Cs interacted with MPK3 and MPK6 and were required for ABA-mediated MPK3/MPK6 inactivation and immune suppression. The bacterial pathogen *Pseudomonas syringae* pv. *tomato* (Pto) DC3000 activates ABA signaling and produces a JA-mimicking phytotoxin, coronatine (COR), that promotes virulence. We found that Pto DC3000 induces *HAI1* through COR-mediated activation of MYC2, a master transcription factor in JA signaling. *HAI1* dephosphorylated MPK3 and MPK6 in vitro and was necessary for COR-mediated suppression of MPK3/MPK6 activation and immunity. Intriguingly, upon ETI activation, *A. thaliana* plants overcame the *HAI1*-dependent virulence of COR by blocking JA signaling. Finally, we showed conservation of induction of *HAI* PP2Cs by ABA and JA in other Brassicaceae species. Taken together, these results suggest that ABA and JA signaling pathways, which are hijacked by the bacterial pathogen, converge on the *HAI* PP2Cs that suppress activation of the immune-associated MAPKs. Also, our data unveil interception of JA-signaling activation as a host counterstrategy against the bacterial suppression of MAPKs during ETI.

MAPK phosphatase | abscisic acid | jasmonate | coronatine | effector-triggered immunity

Plants have an innate immune system consisting of pattern-triggered immunity (PTI) and effector-triggered immunity (ETI) that restricts microbial invasions (1, 2). In PTI, cell-surface-localized pattern recognition receptors detect conserved microbial molecules, termed microbe-associated molecular patterns (MAMPs), leading to a diverse array of immune responses such as a burst of reactive oxygen species, cellular Ca<sup>2+</sup> spikes, activation of MAP kinases (MAPKs), and production of phytohormones (2, 3). Signaling pathways mediated by these immune components form intricate networks that orchestrate defense reactions (2, 4). Pathogens deliver virulence effectors into the plant cell that interfere with immune signaling components, resulting in PTI suppression (5). Recognition of specific effectors by intracellular nucleotide-binding/leucine-rich repeat (NLR) receptors activates ETI, which counteracts pathogen virulence and restricts pathogen growth (6).

The highly interconnected network structure of plant immune signaling pathways provides versatile immune regulation, but is sometimes exploited by pathogens. For example, various strains of the bacterial pathogen *Pseudomonas syringae*, including the model strain *P. syringae* pv. *tomato* (Pto) DC3000, produce the phytotoxin coronatine (COR) that activates jasmonate (JA)

signaling and promotes virulence (7–9). COR is a structural mimic of the biologically active form of JA, (+)-7-iso-jasmonoyl-L-Ile (JA-Ile). Perception of COR or JA-Ile by COI1 triggers proteasome-dependent degradation of JAZ proteins. This releases JAZ-mediated repression of transcription factors, including the master transcription factor MYC2 and its homologs MYC3 and MYC4 (10). Activation of MYC2 by COR contributes to its virulence function of antagonizing salicylic acid (SA)-mediated immunity against *P. syringae* by transcriptionally activating the NAC transcription factors ANAC019, ANAC055, and ANAC072, which repress the SA biosynthesis gene *SID2* and induce the SA catabolism gene *BSMT1* (8). However, COR appears to have another virulence mechanism(s) in addition to SA suppression (8).

The phytohormone abscisic acid (ABA) plays a pivotal role in abiotic stress responses (11). ABA signaling is repressed by clade A protein phosphatases 2C (PP2Cs), which inhibit SNF1-related kinases 2 (SnRK2s) via dephosphorylation. Upon perception, the PYR/PYL/RCAR ABA receptors bind to clade A PP2Cs to release inhibition of SnRK2s. This leads to SnRK2-mediated phosphorylation and activation of a group of AREB/ABF transcription factors, including AREB1, AREB2, and ABF3, thereby regulating genes containing ABA responsive elements (ABREs) in their promoters (12). Interestingly, Pto DC3000 activates ABA biosynthesis and signaling in infected tissues through the actions of proteinaceous effectors delivered by the type III secretion system

## Significance

Pathogens cause disease by deploying virulence effectors that interfere with various host targets, whereas plants counteract pathogen virulence when invoking a potent immunity known as effector-triggered immunity (ETI). Little is known about the mechanism underlying this molecular battle between plant immunity and pathogen virulence. We find that the phytohormones abscisic acid and jasmonate (JA), the signaling pathways of which are often exploited by pathogens, transcriptionally activate a common family of protein phosphatases that suppress immune-associated MAP kinases. We demonstrate that a bacterial pathogen exploits the JA-mediated suppression of MAP kinases by using a JA-mimic, whereas ETI blocks JA signaling to counteract this bacterial virulence. Our results highlight suppression and protection of MAP kinases as a molecular battle between pathogens and plants.

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